

Total No. of Questions : 4]

SEAT No. :

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PD1

[6408]-101

F.E. (Insem)

ENGINEERING MATHEMATICS-II

(2019 Pattern) (Credit System) (Semester-II) (107008)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2 Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of logarithmic tables slide rule, mollier charts, electronic pocket calculator and steam table is allowed.
- 5) Assume suitable data if necessary.

Q1) a) Solve $\frac{dy}{dx} = \frac{1 + y^2 + 3x^2y}{1 - 2xy - x^3}$ [5]

b) Solve $(3xy^2 - y^3)dx + (xy^2 - 2x^2y)dy = 0$ [5]

c) Solve $\sec x \tan y \frac{dy}{dx} + \tan x \sec y = \cos^2 x$ [5]

OR

Q2) a) Solve $\frac{dy}{dx} + \frac{2x}{1+x^2}y = \frac{3x^2}{1+x^2}$ [5]

b) Solve $\frac{dy}{dx} - xy = -y^3e^{-x^2}$ [5]

c) Solve $(x^4e^x - 2mxy^2)dx + 2mx^2y dy = 0$ [5]

Q3) a) A body is heated to 100°C & placed in a room whose temperature is 20°C and cools to 60°C in 5 minutes. How much time is required for it to cool to 40°C? [5]

P.T.O.

- b) A resistance of 150 ohms, an inductance of 0.5 henry are connected in a series with a battery of 30 volts. Find the current in the circuit as a function of t . [5]
- c) A particle of unit mass moves in a horizontal straight line OA with an acceleration $\frac{k}{x^3}$ at a distance 'x' and directed towards 'o'. If initially the particle was at rest at a distance 'a' from o, show that it will be at a distance $\frac{a}{2}$ from 'o' at the end of time $\frac{a^2}{2} \sqrt{\frac{3}{k}}$. [5]

OR

- Q4)** a) Find the orthogonal trajectories of the family of curves $r^2 = a \cos 2\theta$. [5]
- b) A capacitor $c = 0.01$ farad in a series with a resistor $R = 20$ ohms is charged from a battery $E = 10$ volts. Assuming that initially the capacitor is completely uncharged, determine the charge $Q(t)$ and current $I(t)$ in the circuit. [5]
- c) A pipe 10cm in a diameter contains steam at 200°C . It is protected with a covering 5cm thick for which $k=0.12$. If the temperature of outer surface of the covering is 50°C . How much heat is lost per minute from a portion of the pipe 20 metres long? [5]

